InfoBridge: Sprint & Hack Roadmap

Team: The North Remembers · **Project:** DSI 9A (Hyderabad Runners)
Covers: Sprint 2 (27 Sep) → Showcase (20 Nov 2026)

Overview

We build InfoBridge in small steps. Each step adds one capability and is measured on the same synthetic test cases, so we can show improvement at every checkpoint.

Phase	Dates	Version	Focus	Key output
Sprint 2	27 Sep – 10 Oct	v0.1 foundations	Basic architecture and a working baseline	Technical Design Doc, baseline results
Hack 1	10 Oct	v0.1	Connect names across sources; flag conflicts	Demo video, revised doc
Sprint 3	11 – 24 Oct	v0.2	Smarter extraction, knowledge graph, more test cases	All components in place, test results
Sprint 4	25 Oct – 7 Nov	v0.9	Testing, edge cases, robustness, open data	Evaluation and failure analysis
Hack 2	7 Nov	v1.0	Polish and final evaluation	Final Technical Design Doc
Sprint 5	8 – 14 Nov	v1.0 final	Package and defend	Slides, poster, video, final repo

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Sprint 2: Basic architecture & implementation

Goal: a first working version that answers a question from several sources with citations, plus a baseline score to improve on.

Architecture (v0.1)

```
Source files:  CSV · Excel · PDF · WhatsApp export · Email
       │
       ▼
[1] Ingestion        split each file into small chunks (row / message /
       │             paragraph), each tagged with its exact location
       ▼
[2] Vector store     embed the chunks; find the ones most relevant to the
       │             question  (ChromaDB + all-MiniLM-L6-v2)
       ▼
[3] Synthesis        LLM answers using only the retrieved chunks; every
       │             claim must cite a source  (Gemini)
       ▼
[4] Citation check   flag claims with missing or invalid citations
       │
       ▼
Answer:  verdict + cited claims + sources list
         (or "not enough evidence" + where to look)
```

Later sprints add these blocks between ingestion and synthesis: **fact extraction → entity resolution → conflict detection → knowledge graph.**

What we build in Sprint 2

Part	Tech	What it does
Ingestion	pandas, PyMuPDF	Reads CSV, Excel, PDF, WhatsApp exports and emails; splits them into chunks with an exact location (e.g. `members.csv` row 5)
Vector store	ChromaDB + all-MiniLM-L6-v2	Finds the chunks most relevant to the question
Synthesis	Gemini (free tier)	Writes the answer using only the retrieved chunks; every claim must cite a source
Citation check	Python	Flags claims that cite nothing, or cite a source that wasn't retrieved
Fallback	Python	If nothing relevant is found, says so and points to the closest sources instead of guessing
Synthetic data	Hand-written	3 test cases (simple / medium / conflict), each with an answer key
Evaluation	Python script	Scores answers against the answer key

Not in Sprint 2: entity resolution, conflict detection, knowledge graph. The baseline is supposed to fail where these are needed, which shows why we need them.

Test cases

Case	Sources	Question	What makes it hard
1 – Simple	Events sheet, email	Details of the 25 Oct long run	Nothing. This is the control case
2 – Medium	Volunteer roster, WhatsApp, first-aid PDF	Who's at Aid Station 3, and are they trained?	Same people with different names (*Srini* = *Srinivas Rao Kumar*); a look-alike (*Srinivas Reddy*)
3 – Conflict	Members sheet, registrations, WhatsApp, rules PDF	Is Ravi Sharma eligible for the training programme?	*Ravi Sharma / R. Sharma / Ravi S.*; look-alike *Ravi Shankar*; 2 contradictions between sources

Evaluation criteria

Metric	Question it answers
Retrieval recall	Did we find the chunks that hold the answer?
Fact recall	Did the answer include all the key facts?
Citation validity	Does every citation point to a real source chunk?
Conflict recall	Did we flag the planted contradictions?
Verdict accuracy	Was the YES/NO correct?
Latency	How long did it take?

Baseline result (retrieval only)

Case	Retrieval recall	Fact recall	Citation validity	Conflict recall
1 – Simple	1.00	1.00	1.00	–
2 – Medium	0.67	0.50	1.00	–
3 – Conflict	0.57	0.14	1.00	0.00

- **Worked:** every citation points to an exact location; the simple case is answered completely.

- **Didn't work:** when names differ across sources, the right chunks aren't found. Ravi Sharma's own membership row ranks 20th of 29, below the look-alike Ravi Shankar. Conflicts are never flagged.
- **To do:** add the Gemini baseline numbers once an API key is set up.

Sprint 2 deliverables

☑ Basic pipeline running end to end (code on GitHub)

☑ 3 synthetic test cases with answer keys

☑ Baseline results: what worked and what didn't

☐ Gemini baseline run

☐ Technical Design Doc submitted (Moodle)

Hack 1 (10 Oct): Make it work, find what breaks

Goal: fix the two biggest baseline failures and demo it live.

1. **Simple entity resolution:** treat records as the same person if the phone or email matches, or if the names are compatible (*R. Sharma* ~ *Ravi Sharma*, *Srini* ~ *Srinivas*). Never merge different surnames (*Sharma* ≠ *Shankar*).
2. **Simple conflict detection:** compare values for the same person across sources (dates, statuses) and flag any differences, using spreadsheet columns plus dates found in text.
3. **Show conflicts in the answer:** list each conflict with both values and their sources.
4. **Re-run the evaluation** and compare with the baseline.

Deliverables: revised Technical Design Doc, demo video (question → answer → evidence → conflict), PoC v0.1 on GitHub, list of issues and questions for instructors.

Sprint 3 (11 – 24 Oct): Build it out → v0.2

Goal: all 7 components in place, clearly better than Hack 1.

- Act on Hack 1 feedback; fix the worst failures first.
- **LLM fact extraction** from chats, PDFs and emails (not just spreadsheet columns).
- **Knowledge graph** (NetworkX): people, events and their facts, each linked to its source.
- **Better retrieval:** fetch every chunk about the person asked about, not just the most similar text.
- **Full evidence trail:** source → extracted fact → matched entity → claim.
- **More test cases:** 5–6 in total, including one written by a teammate who didn't build the matcher.

Deliverables: PoC v0.2 on GitHub, test spec and results, known limitations.

Sprint 4 (25 Oct – 7 Nov): Validate & harden → v0.9

Goal: credible, tested, stable.

- **Systematic testing:** unit tests per component plus end-to-end tests on all cases.
- **Edge cases:** empty files, questions with no answer in the sources, heavily conflicting sources, near-identical names.
- **Robustness:** handle LLM errors and rate limits; cache results; fail with a clear message rather than silently.

- **One real open-data scenario** (OGD India / WHO / OpenAlex) to show the approach works beyond our own test data.
- **Better scoring:** a manual or LLM-based check of answer quality, in addition to keyword matching.

Deliverables: revised Technical Design Doc (evaluation, failure analysis, limitations), PoC v0.9, demo plan for Hack 2.

Hack 2 (7 Nov): Complete & polish → v1.0

- Fix the remaining high-priority issues.
- Make outputs clean and consistent; make the demo run reliably.
- Final evaluation run; a results table comparing every version (v0.1 → v1.0).
- Prepare a backup recording in case the live demo fails.

Deliverables: final Technical Design Doc, PoC v1.0 on GitHub.

Sprint 5 (8 – 14 Nov): Package & defend

- Clean the code; write the README (setup, how to run, architecture).
- Final versions of the Functional Scope Doc, Design Doc and Technical Design Doc.
- **Slides, poster and demo video:** problem → solution → technology → results.
- Review with NGOs (table discussions).
- Prepare for likely questions:
  - "How is this different from ChatGPT?"
  - "Would it work on real NGO data?"
  - "What would production need?"
  - "Which other NGOs could use it?"

Then: Demo 2 (14 Nov) · Final assessment (18 Nov) · Showcase (20 Nov).

How the metrics should improve

Metric	v0.1 baseline	Hack 1	v0.2	v1.0
Retrieval recall (case 3)	0.57	–	≥ 0.90	≥ 0.90
Conflict recall	0.00	≥ 0.50	≥ 0.90	≥ 0.90
Entity resolution F1	–	measured	≥ 0.85	≥ 0.90
Citation validity	1.00	1.00	1.00	1.00
Verdict accuracy	TBD	–	1.00	1.00

These are targets on our synthetic cases only. They don't show that the system would work on real NGO data; understanding that gap is part of the project.